In-memory compression of Awkward Arrays with Coffea

Peter Elmer[1], **Peter Fackeldey[1]**, Iason Krommydas[2],
Princeton University[1], Rice University[2]

Memory Limits Kill Your Jobs

Chunk too big ⟶ crash. Chunk too small ⟶ slow.

Limited memory forces tiny chunks, causing:

- Task scheduling overhead
- Inefficient use of vectorized kernels
- Python overhead

Reduced memory footprint ⟶ more robust & faster analysis

How Awkward Arrays Are Stored in Memory

Nested data (e.g. per-event jet lists) are stored as flat 1D buffers:

- **Offsets:** where each event starts/ends
- **Values:** all data flattened into one array

100k events ⟶ just 2 contiguous buffers (not 100k Python lists!).

Each buffer has a **unique key**. This maps naturally to a key-value store for compression and lazy lookup.

User View

Jet-momenta for variable number of Jets
Event 0: **[45.2, 32.1, 12.5]**
Event 1: **[67.8]**
Event 2: **[23.0, 91.4]**

1D Buffers

In-memory Layout of Awkward Arrays
Offsets: **[0, 3, 4, 6]**
Values: **[45.2, 32.1, 12.5, 67.8, 23.0, 91.4]**

Key-Value Store NEW!

Key-Value Cache Store for 1D Buffers
`offsets/jets/pt` ⟶ Offsets
`data/jets/pt` ⟶ Values

Benefits of a Key-Value Store for Buffers:

- Full control over caching strategy
- Only accessed buffers are loaded into the key-value store (VirtualArrays)
- Transparent to users

Solution: `BufferCache` for Awkward Arrays

New in coffea: plug any key-value store (`BufferCache`) underneath your arrays.

```python
from coffea.nanoevents import NanoEventsFactory
from coffea.nanoevents.mapping import BufferCache

buffer_cache = BufferCache(...)

factory = NanoEventsFactory.from_root(
  {"path/to/nanoaod.root": "Events"},
  mode="virtual",
  buffer_cache=buffer_cache,
)
```

Three strategies below:

1. In-Memory Compression

Compress buffers before caching. Decompress only on access ⟶ larger chunks, same memory limit.

Example with Blosc:

```python
from coffea.nanoevents.mapping import BufferCache
from numcodecs import Blosc

buffer_cache = BufferCache(
  cache={},
  codec=Blosc("zstd", clevel=1, shuffle=Blosc.BITSHUFFLE)
)
```

2. On-Disk Caching

Spill buffers to disk. Bypasses memory limits entirely — at the cost of I/O latency.

```python
from coffea.nanoevents.mapping import BufferCache
from numcodecs import Blosc; import zict

buffer_cache = BufferCache(
  cache=zict.File("cache_dir"),
  codec=Blosc("zstd", clevel=1, shuffle=Blosc.BITSHUFFLE)
)
```

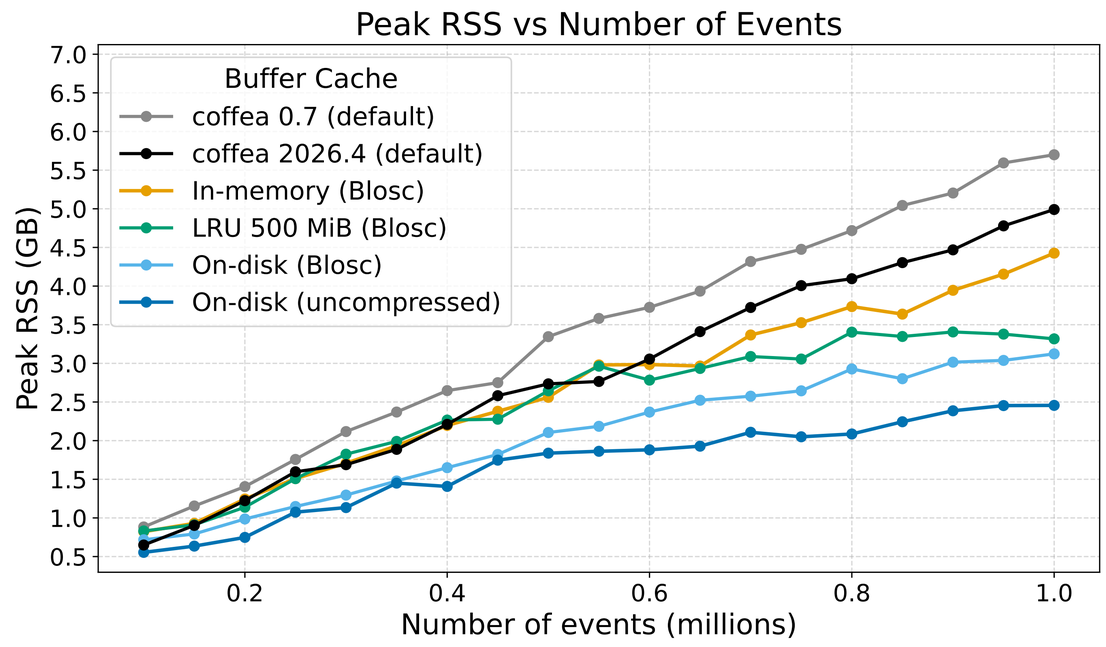
3. Custom / Tiered Caches

Any `MutableMapping[str, bytes]` works — e.g. tiered caches that spill to disk once a memory threshold is reached, or distributed caches for shared access between users.

Benchmarks

Comparing memory use for a typical coffea analysis, using the *Analysis Grand Challenge* with a CMS open data $t\bar{t}$ NanoAOD sample:

- No cache (default): `coffea 0.7` and `coffea 2026.4`
- In-memory (`Blosc`)
- On-disk (`zict.File`, SSD-backed)
- Tiered LRU (`zict.LRU`)



Results

- Peak RSS **reduced by up to ≈2×** with on-disk caching
- In-memory compression also reduces peak RSS vs default, especially at higher chunk sizes
- Runtime overhead ≈1..30% (depends on cache strategy and chunk size)
  - Can be mitigated by larger chunks and codec tuning
  - On-disk performance is *highly sensitive to disk bandwidth*
    ⟶ understand your setup!

⟶ Try `coffea>=2026.4` if memory limits slow you down!

Acknowledgements: This work was supported by the National Science Foundation under Cooperative Agreement PHY-2323298 and grant OAC-2103945

Contact: peter.fackeldey@cern.ch